ENERGY AUDIT REPORT

Residential Energy Audit and Energy Conservation Measures

Digital Assignment 1 (DA1) - Thermodynamics

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Audit subject	Residential building (3-BHK, family of four)
Audit type	Preliminary energy audit (BEE classification)

Executive Summary

The audited home consumes about 6317 kWh a year at a cost near Rs 37,900 (17.3 kWh/day). Air conditioning, ceiling fans and lighting account for most of it. Eight measures, from a zero-cost setpoint change to BLDC fans and an inverter AC, cut consumption by 42% (to 3663 kWh), save Rs 15,922 a year and avoid 2.2 tonnes of CO_2 a year. The package costs Rs 68,200 with a simple payback of 4.3 years and a positive ten-year NPV at 8%.

1. Introduction and Objectives

Energy is one of the largest recurring costs of a household, and also one of the most controllable. An energy audit is the verification, monitoring and analysis of energy use, followed by a technical report with recommendations, cost-benefit analysis and an action plan (Energy Conservation Act, 2001). This report applies the preliminary-audit approach described by the Bureau of Energy Efficiency to a residential building: establish consumption, find the largest end uses, and identify no-cost and low-cost savings before proposing capital measures.

Objectives

(i) Build an inventory of electrical loads and estimate annual consumption and bill. (ii) Identify where energy is wasted in lighting, fans, water pumping, air conditioning and other devices. (iii) Assess the building envelope, windows and ventilation. (iv) Propose energy conservation measures (ECMs) with savings and simple payback. (v) Connect each recommendation to the underlying thermodynamic principle.

2. Methodology

The audit followed the three phases in the BEE guide. **Pre-audit:** walk-through of all rooms, listing every load with its rated power and daily usage, and collecting the electricity bills. **Audit:** energy consumption was computed per appliance as $E = P \times t$, aggregated by month, and compared against the bill. **Post-audit:** measures were ranked by cost and payback and grouped into No-investment (A), Low-investment (B) and High-investment (C) classes, as in BEE Table 3.3. Following the BEE data-collection hint, nothing was estimated that could be calculated, and nothing calculated that could be measured: the plug-in wattmeter and the nameplate are the preferred sources, and a lux meter and a thermometer are used for lighting and AC checks.

Basis of data. Appliance ratings and hours of use are taken for a typical three-bedroom home with four occupants. The blended tariff is taken as Rs 6.00 per kWh (energy plus fixed charges).

3. Thermodynamic Basis

First law. For a building taken as a control volume, electrical energy supplied is converted to heat inside the space; the audit is an energy balance, $E_{in} = E_{useful} + E_{losses}$. A device that wastes less delivers the same service for less input. **Second law.** Electrical energy is high-grade work; using it to make low-grade heat (geysers, resistance heaters) destroys availability, so such loads are the first candidates for timers, lower setpoints or solar substitution. **Heat transfer.** Heat gain through the envelope is $Q = U\,A\,\Delta T$. Reducing U (insulation, better glazing) or A (shading) reduces the cooling load directly. **Refrigeration cycle.** An air conditioner removes Q_L using work W with $COP = Q_L/W$, bounded by the Carnot value $T_L/(T_H - T_L)$. Raising the setpoint from 22 C to 26 C narrows the temperature lift, raising COP and cutting compressor work. **Fan affinity laws.** Power varies with the cube of speed, which is why variable-speed (BLDC, VFD) drives save so much in fans and pumps.

4. Load Inventory and Energy Bill Statistics

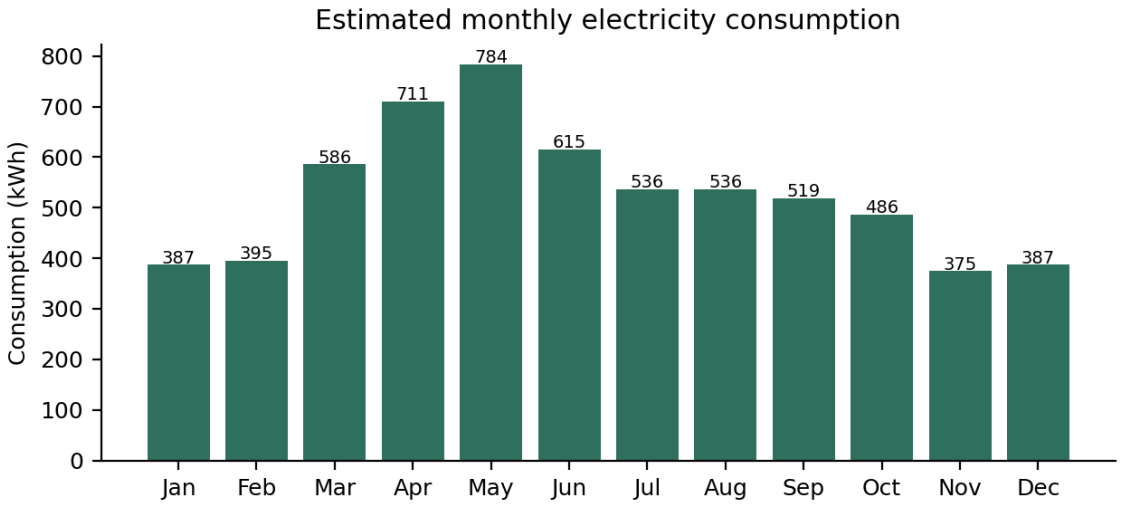
Table 1 lists every load found during the walk-through with existing rating, usage and the annual energy computed from E = P x t.

Appliance	Qty	Rated W	Hrs/day	kWh/year	Rs/year
Tube light T8 + choke	6	50	6	657	3,942
CFL lamp	5	15	5	137	821
Ceiling fan	5	75	10	1369	8,212
Refrigerator 250 L	1	67	24	587	3,522
Water pump 0.5 HP	1	600	1.5	328	1,971
Geyser 2 kW	1	2000	0.5	365	2,190
Television	1	100	4	146	876
Washing machine	1	500	0.7	128	766
Mixer / grinder	1	500	0.3	55	328
Laptop, chargers, WiFi	1	70	8	204	1,226
Split AC 1.5 ton (3-star)	1	1600	seasonal (see Table 2)	2341	14,045
Total				**6317**	**37,900**

Table 1: Connected load and annual energy (existing condition).

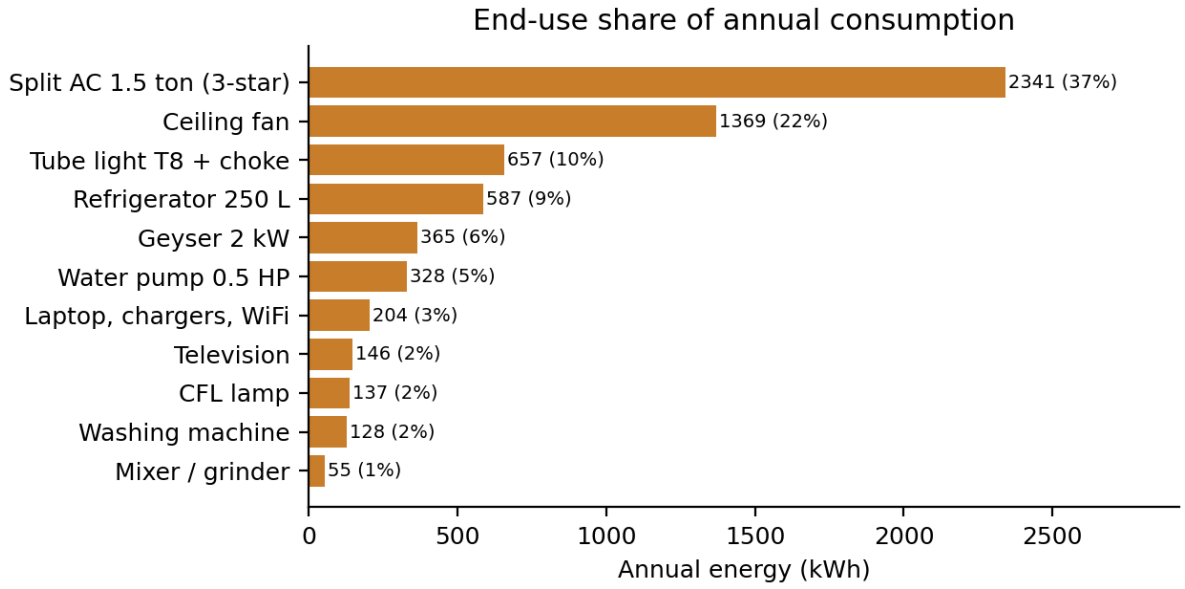
Month	Days	AC hrs/day	Units (kWh)	Bill (Rs)
Jan	31	1	387	2,324
Feb	28	2	395	2,368
Mar	31	5	586	3,514
Apr	30	8	711	4,265
May	31	9	784	4,704
Jun	30	6	615	3,689
Jul	31	4	536	3,216
Aug	31	4	536	3,216
Sep	30	4	519	3,113
Oct	31	3	486	2,919
Nov	30	1	375	2,249
Dec	31	1	387	2,324
Year	365		**6317**	**37,900**

Table 2: Month-wise consumption and bill.



Observations. Annual consumption is about 6317 kWh costing roughly Rs 37,900, an average of 526 units and Rs 3,158 per month. Consumption peaks in May (784 kWh) and is lowest in Nov (375 kWh); the 2.1x swing is driven almost entirely by air-conditioner use, which is a clear signal that the cooling load is the main lever. Specific consumption is 17.3 kWh per day, or 1579 kWh per person per year (benchmark parameter, internal).

5. Where the Energy Goes



The three largest end uses are Split AC 1.5 ton (3-star) (37%), Ceiling fan (22%) and Tube light T8 + choke (10%), together accounting for 69% of the bill. Effort is therefore concentrated on these first, in line with the BEE advice to start with the most likely and easiest areas.

6. Observations by Area

Lighting. Older T8 fluorescent tubes with magnetic chokes draw about 50 W including loss, and CFLs add to this. LED replacements give equal or better lux at 18 W and 9 W. Light levels should be checked with a lux meter so that no room is over-lit; task lighting at the study table and kitchen counter is better than raising general light levels.

Fans. Conventional induction fans draw about 75 W at full speed with poor efficiency at low speed. BLDC fans deliver the same airflow at about 28 W and keep efficiency at part load. Fans also run when rooms are empty; occupancy habits matter.

Water pumping. A 0.5 HP pump is used for roughly 1.5 hours per day. Common losses are throttling at the valve, a leaking foot valve causing repeated priming, and an oversized pipe-friction path. A float switch or timer stops overflow and dry running, and a correctly sized, higher-efficiency pump reduces input power. Pump power $P = \rho\, g\, Q\, H / \eta$, so reducing head losses or increasing η reduces power directly.

Air conditioning. The AC is the largest seasonal load. A 3-star fixed-speed unit cycles on and off and runs at lower COP at high ambient temperature. Observed practice of 22-23 C setpoints raises consumption; a 1 C rise in setpoint cuts compressor work by about 3% on a Carnot basis (field estimates go up to 6%), so a conservative 8% is assumed for a 2-3 C rise. Dirty filters, blocked outdoor unit airflow and a leaky room multiply the load.

Refrigerator and other appliances. The refrigerator is on all day; loss comes from worn door gaskets, warm placement near the stove and poor clearance behind the condenser. Standby power from TV, set-top box, chargers and the router is small per device but runs 24 hours.

Windows, envelope and ventilation. Windows are the weakest thermal element: single-pane glazing has a U-value above about 5 W/m^{2}K against walls near 2 W/m^{2}K. West and south windows without external shading admit direct solar gain in the afternoon, forcing the AC to work harder. Gaps around doors and window frames allow infiltration of hot humid air. At the same time, adequate ventilation is needed for air

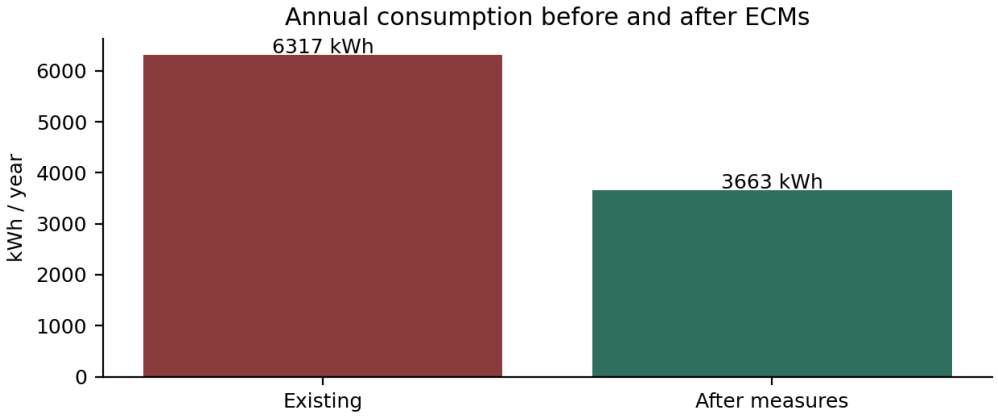
quality, so sealing must be paired with working exhaust fans in kitchen and bath, vented to the outside, and cross ventilation in the evening.

7. Energy Conservation Measures and Savings

Each measure was evaluated as: annual saving = (existing kWh - proposed kWh); money saved = kWh x tariff; simple payback = investment / annual money saved.

#	Measure	Saving kWh/yr	Saving Rs/yr	Investment Rs	Payback (yr)	Class
1	Replace TV at end of life (60 W LED)	58	350	0	Immediate	A - No/very low
2	Raise AC setpoint to 24-26 C (behaviour)	123	737	0	Immediate	A - No/very low
3	Replace T8 tubes and CFLs with LED	475	2,851	2,700	0.9	B - Low
4	Geyser timer and 50 C setpoint	91	548	800	1.5	A - No/very low
5	Replace ceiling fans with BLDC fans	858	5,146	16,000	3.1	C - High
6	Smart power strip (standby cut)	58	350	1,500	4.3	B - Low
7	Pump float switch/timer, fix leaks	44	263	1,200	4.6	B - Low
8	Window film, curtains, sealing (-10% AC)	141	848	4,000	4.7	B - Low
9	Inverter 5-star AC (hardware)	805	4,828	42,000	8.7	C - High
	Total	**2654**	**15,922**	**68,200**	**4.3**	

Table 3: Summary of energy saving recommendations (format per BEE Tables 3.2 and 3.3). The TV row has zero investment because it is counted as a replacement already due.



Result. Implementing all measures lowers annual consumption from 6317 to 3663 kWh, a reduction of 42%, and cuts the yearly bill by about Rs 15,922. The total outlay of Rs 68,200 has a combined simple payback of 4.3 years. The class A and B measures alone cost Rs 10,200 and save Rs 5,948 per year.

8. Housekeeping and Design Recommendations (no or low cost)

Lighting and fans: switch off when leaving a room; clean tubes and diffusers, since dust can cut light output by 10-15%; use natural daylight near windows; use the regulator or lower speed at night. **AC:** set 24-26 C with the fan on auto; clean filters monthly; keep doors and windows shut while running; shade the outdoor unit and keep clearance around it; use timer or sleep mode. **Pump:** install a float switch on the overhead tank, fix leaks, run the pump in off-peak hours where time-of-day tariff applies, and keep the suction line short and tight. **Geyser:** switch on only 15 minutes before use, set the thermostat to 50 C, insulate hot pipes. **Refrigerator:** keep 5-10 cm clearance at the back, check the door seal using the paper test, do not load hot food, defrost regularly. **Standby:** switch off at the plug with a power strip.

Windows and envelope design. Add external shading (chajja, louvres or plants) on west and south windows; use reflective film or heavy curtains on unshaded glass; seal gaps with weatherstrip and sealant; for future renovation choose double-glazed or low-emissivity glass with low U-value and low SHGC, and consider cool-roof paint or roof insulation, which reduce $Q = UA\Delta T$ through the largest exposed surface.

Ventilation. Provide cross ventilation by opening windows on opposite sides in the evening; ensure kitchen and bathroom exhaust fans work and discharge outdoors; use stack effect through a high-level opening; prefer a ceiling fan to extend the comfort range so the AC setpoint can rise by 2 C.

Monitoring. Record the meter reading on the 1st of every month, compute specific consumption (kWh/day) and track it, as recommended under BEE benchmarking. Sub-metering the AC and pump would confirm the estimated savings.

9. Economic and Environmental Evaluation

Simple payback ignores the time value of money, so each measure is also assessed by net present value over 10 years at 8% (annuity factor 6.71), with the tariff held constant. CO_2 is estimated with a grid factor of 0.82 kg/kWh (assumption). Most measures show a positive NPV, but the inverter AC does not pay back on energy savings alone within the project life, so it should be bought when the existing unit is due for replacement, after the low-cost measures.

Measure	Rs saved/yr	Investment Rs	Payback (yr)	NPV Rs	CO2 avoided kg/yr
Replace ceiling fans with BLDC fans	5,146	16,000	3.1	18,533	703
Inverter 5-star AC (hardware)	4,828	42,000	8.7	-9,604	660
Replace T8 tubes and CFLs with LED	2,851	2,700	0.9	16,433	390
Window film, curtains, sealing (-10% AC)	848	4,000	4.7	1,690	116
Raise AC setpoint to 24-26 C (behaviour)	737	0	Immediate	4,948	101
Geyser timer and 50 C setpoint	548	800	1.5	2,874	75
Smart power strip (standby cut)	350	1,500	4.3	851	48
Replace TV at end of life (60 W LED)	350	0	Immediate	2,351	48
Pump float switch/timer, fix leaks	263	1,200	4.6	563	36
Total	**15,922**	**68,200**	**4.3**	**38,639**	**2176**

Table 4: Economic and environmental evaluation.

10. Thermodynamic Checks

Window solar gain. For a 1.5 m^2 west window with irradiance 600 W/m^2, bare glass (SHGC 0.86) admits 774 W; a reflective film (SHGC 0.45) admits 405 W. The 369 W cut in cooling load needs 123 W less electrical input at COP 3.0, about 98 kWh a year per window for 4 h/day over 200 days. **AC setpoint.** Treating the AC as a reversible refrigerator between an evaporator at setpoint minus 10 K and a condenser at 50 C (35 C outdoor plus 15 K), the ideal COP rises with setpoint, so compressor work falls (Table 5).

Setpoint (C)	Carnot COP	Relative compressor work
20	7.08	100%
22	7.50	94%
24	7.98	89%
26	8.50	83%

Table 5: Carnot-based effect of setpoint (illustrative model; real units reach roughly 40% of Carnot COP).

Plant energy performance. After implementation, BEE's PEP = (reference-year equivalent - current use) / reference-year equivalent x 100 should be tracked monthly. With the same occupancy (production factor 1), the target PEP is about 42%.

11. Conclusion

The audit shows an annual consumption of about 6317 kWh and a bill near Rs 37,900, dominated by air conditioning, fans and lighting. A package of lighting, fan, pump, water heating, standby and cooling-load measures can reduce consumption by about 42% with the quickest returns from LED lighting, timers and housekeeping. The same principles used in the BEE industrial audit, namely measure, benchmark, match energy use to requirement, maximise system efficiency and then consider substitution, apply directly at household scale and rest on the first and second laws of thermodynamics.

References

1. Bureau of Energy Efficiency, Chapter 3: Energy Management and Audit (Guide for National Certification Examination).
2. Energy Management Centre Kerala, Energy Audit Manual.
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4. Energy Conservation Act, 2001, Government of India.
5. Y. A. Cengel and M. A. Boles, Thermodynamics: An Engineering Approach.